

Appendix A

Overview of Grade Bands and AI Literacy Components Associated with the Reviewed Studies

Reviewed Research	Brief Description	K–6	7–8	9–12	Understanding Foundational AI Concepts	Creating AI Artifacts	Interacting With AI Agents	Developing Awareness of AI Ethics	Understanding Human-AI Relationships
Aung et al., 2022	This study describes a web-based platform (AIThaiGen) that talks about enabling junior high students to test AI models with real hardware and reports significant gains in AI concept understanding and positive attitudes.		X		X	X		X	
Bellas et al., 2023	This study describes a two-year, smartphone-based AI curriculum developed and tested across European high schools and talks about its feasibility for near-term school adoption.			X	X	X		X	X
Cai, 2024	This study describes Shanghai primary students' views of AI as both companion and privacy risk and talks about their expectations for more capable, emotionally aware future AI.	X			X				X
Chai et al., 2021	This study discusses the results of a survey study of 682 students about how self-efficacy, readiness, social good, and literacy predict intentions to learn AI.	X			X		X		X
Chiu et al., 2021	This study describes the co-creation of a secondary AI curriculum and talks about improvements in student competence and teacher capacity.		X	X	X		X	X	X
Coenraad, 2022	This study describes co-design sessions with Black youth and talks about their awareness of visible technological and algorithmic bias and the need for language to discuss less visible forms.	X	X		X	X		X	X
Dai et al., 2024a	This study describes an analogy-based teaching approach centered on human–AI comparison and talks about its superior effects on AI knowledge, skills, and ethics.	X			X				X

Reviewed Research	Brief Description	K–6	7–8	9–12	Understanding Foundational AI Concepts	Creating AI Artifacts	Interacting With AI Agents	Developing Awareness of AI Ethics	Understanding Human-AI Relationships
Dai et al., 2024b	This study explores the effects of a pedagogical approach that aims to improve upper-elementary school students' scientific understanding of AI through analogical teaching and embodied interaction.	X			X	X	X		
Druga et al., 2019	This study describes cross-country workshops with children and talks about how socioeconomic and cultural contexts shape AI perceptions and collaboration.	X	X		X	X	X	X	
Eguchi et al., 2021	This study describes a Japanese K–12 AI initiative and talks about culturally responsive strategies to promote ethical awareness and contextualized learning.		X		X		X		X
Hsu et al., 2022	This study examines the effects of an AI curriculum designed for students to learn the application of conversational AI on a block-based programming platform for secondary schools.		X		X	X			
Hu et al., 2024	This study explores how students interact within groups when learning AI through hands-on activities in the four group learning types.	X				X			
Huang, 2021	This study describes a fundamental-stage AI curriculum and talks about its positive effects on six key student sub-competencies and areas for improvement.		X		X	X	X		
Jiang et al., 2022	This study describes how students build text-classification models and talks about their feature engineering, handling of noisy data, and reasoning about model learning.			X	X	X		X	
Jiang et al., 2023a	This study describes 28 students' development of text-based machine learning models and talks about nine modeling practices and emerging perspectives on automated decisions.			X	X			X	

Reviewed Research	Brief Description	K–6	7–8	9–12	Understanding Foundational AI Concepts	Creating AI Artifacts	Interacting With AI Agents	Developing Awareness of AI Ethics	Understanding Human-AI Relationships
Jiang et al., 2023b	This study examines the engagement of students from a journalism class in refining machine learning models iteratively for classifying negative and positive reviews of ice cream stores.			X	X			X	
Kim & Kwon, 2024	This study describes an elementary AI curriculum using tangible tools and talks about gains in AI knowledge, perceptions, and intention to learn.	X			X	X	X	X	
Kim & Lee, 2024	This study describes a survey of 1,677 middle schoolers and talks about how gender, experiences, and education shape attitudes toward AI, with education mitigating disparities.		X		X				
Kim et al., 2023	This study describes six common naive AI conceptions found during a summer camp and talks about how these ideas evolved through targeted activities.	X	X		X	X		X	
Kong et al., 2023	This study describes a 34-hour program on AI literacy education and talks about improved concepts and ethical awareness among participants, while noting persistent challenges in deep ethical comprehension.			X	X	X			X
Lee et al., 2021	This study describes a 30-hour online AI workshop for underrepresented middle school students and talks about how it supported engagement, conceptual learning, and career-oriented views of AI.		X		X			X	X
Lin et al., 2024	This study describes a structural equation model of 509 Chinese students and talks about how AI literacy and deep learning approaches strengthen computational thinking efficacy.			X	X	X	X		X
Ng et al., 2024a	This study describes a project-based maker program where students built AI recycling bins and talks about gains in motivation, collaboration, and literacy across cognitive domains.		X		X	X	X	X	

Reviewed Research	Brief Description	K–6	7–8	9–12	Understanding Foundational AI Concepts	Creating AI Artifacts	Interacting With AI Agents	Developing Awareness of AI Ethics	Understanding Human-AI Relationships
Ng et al., 2024b	This study describes the development of a 32-item AI literacy instrument and talks about its reliability and validity across affective, behavioral, cognitive, and ethical dimensions.	X	X	X	X	X		X	X
Ottenbreit-Leftwich et al., 2023	This study describes 4th and 5th graders' everyday conceptions of AI alongside teacher reflections and talks about entry points for co-designing age-appropriate AI curricula.	X			X	X			X
Park & Kwon, 2024	This study describes the design and testing of a 16-hour AI program in the free semester system and talks about its effectiveness in boosting technology interest, career aspirations, and AI competency.		X		X	X	X		X
Sing et al., 2022	This study describes a Theory of Planned Behavior-based structural model of AI learning intention and talks about how readiness, optimism, and social-good perceptions moderate student attitudes.			X	X	X	X		X
Su & Yang, 2024	This study describes a Hong Kong kindergarten AI program (AI4KG) and talks about increased robot perceptions but limited effects on science/engineering attitudes, with positive parent feedback.	X			X	X			
Tena et al., 2022	This study describes an autoethnographic case of a Latinx high school student and talks about how her personal background shaped socially just applications of AI.			X	X	X		X	X
Utami & Winarni, 2023	This study describes Indonesian students' perceptions of AI writing tools and talks about their benefits for planning and drafting, alongside challenges with Indonesian-language editing and feature gaps.			X			X		

Reviewed Research	Brief Description	K–6	7–8	9–12	Understanding Foundational AI Concepts	Creating AI Artifacts	Interacting With AI Agents	Developing Awareness of AI Ethics	Understanding Human-AI Relationships
Williams et al., 2023	This study describes three U.S. middle school AI curricula that integrated ethics with active learning and talks about their effectiveness in engaging students across diverse backgrounds.		X		X	X	X	X	X
Xia et al., 2022	This study explores how Self-determination theory (SDT)-based needs support by teachers and student attributes (gender and achievement level) affect AI learning at secondary school level.			X	X	X			X
Yang et al., 2024	This study describes a 6-week Hong Kong kindergarten program using intelligent agents and talks about how embodied activities helped 5-year-olds grasp AI concepts.	X			X		X		
Yetisensoy & Rapoport, 2023	This study describes the intersection of AI literacy and social studies and talks about how hands-on lesson plans can position AI as part of active citizenship education.	X					X		
Zhan et al., 2022	This study describes an experiment comparing two- and three-student groups and talks about how smaller groups improved motivation, collaboration, and cognitive load management in AI learning.			X	X	X			
Zhang et al., 2023	This study describes a middle school AI workshop integrating ethics, technical learning, and career futures and talks about students' improved ability to see AI as both technical and sociocultural.		X		X	X	X	X	X